

BRIDGE: Balancing Representations by Identifying and Generating Underrepresented Data

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Abstract

Self supervised pretraining inherits the sampling bias of its source data. Most remedies for long tailed self supervised learning are model centric. They modify the loss, the schedule, or the optimization dynamics while leaving the training distribution fixed. We present BRIDGE, a data centric alternative that alternates pretraining, mean k NN OOD scoring, and targeted image generation with a frozen diffusion model. The final method replaces a simpler extreme tail selector with a histogram based mode window that targets the modal shoulder of the OOD distribution and avoids the noisy far tail. Across ImageNet-100-LT, CIFAR-10-LT, CIFAR-100-LT, PASS, and DiffusionDB source pretraining, BRIDGE achieves the best average linear probe transfer among the evaluated methods. On ImageNet-100-LT source pretraining, it improves the seven task average over imbalanced SimCLR by 3.05 points, and across the five source regimes its gain over source matched SimCLR ranges from 2.41 to 4.89 points. Ablations show that Stable Diffusion 3 is the strongest generator, SimCLR is the strongest SSL objective inside the repair loop, and the mode window selector is a better replacement for pure top tail sampling.

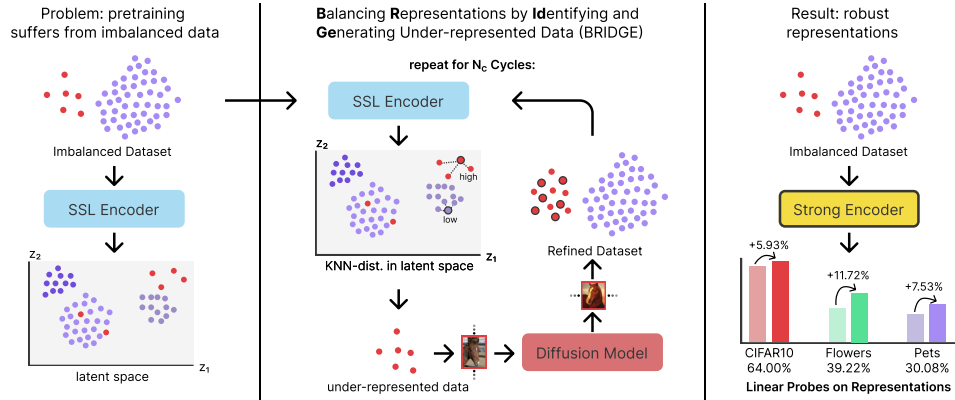


Figure 1: Conceptual overview of BRIDGE. The encoder is trained on the current unlabeled source set, sparse points in representation space are identified by mean k NN distance, and a frozen image to image diffusion model generates additional views for the selected region. The final method uses the mode window selector from Section 3.2 rather than a pure extreme tail rule.

1 Introduction

Self supervised learning has become a standard way to learn transferable visual representations from unlabeled images. Strong methods now span contrastive learning, masked prediction, and self distillation [Chen et al., 2020, He et al., 2020, 2022, Caron et al., 2021, Oquab et al., 2024]. Their success, however, still depends on the geometry of the source data. When the source corpus is long tailed or otherwise skewed, the learned representation devotes most of its capacity to dense regions and leaves sparse regions weakly modeled. This is a data problem as much as it is an optimization problem, and it can reduce downstream transfer on both in domain and fine grained targets.

Most prior attempts to address long tailed SSL are model centric. Temperature Schedules reshapes the optimization trajectory of contrastive learning over time [Kukleva et al., 2023]. SDCLR introduces a self competitor to emphasize harder and rarer samples [Jiang et al., 2021]. Other work adds external OOD data or frequency aware training heuristics to the SSL loop [Bai et al., 2023, Lin et al., 2023]. These methods are important baselines, but they still treat the source distribution as mostly fixed.

We study the complementary direction and ask whether the source dataset itself can be repaired while SSL training proceeds. We propose BRIDGE, a cyclic data repair loop shown in Figure 1. After each pretraining stage, we embed the current training set, score each image by mean k NN distance, select a sparse but still semantically recoverable region of the representation space, and generate new training images for that region with a frozen diffusion model. The next cycle trains on the repaired dataset. The encoder therefore reveals where the data manifold is thin, and the generated images densify those regions before the next round of learning.

Two aspects of the final design are central to the present study. First, the main method no longer selects the most extreme OOD samples. Instead it builds a histogram over OOD distances and allocates a fixed sampling budget around the modal region of that histogram. This skips the super OOD tail, which often contains noisy or weakly useful images. The pure tail rule is retained as an explicit ablation. Second, the empirical scope is substantially broader. In addition to ImageNet-100-LT, CIFAR-10-LT, and CIFAR-100-LT, we evaluate web scale unlabeled source data from PASS and DiffusionDB. BRIDGE is best on the seven task average in all five source regimes that we study.

The paper makes three contributions. First, it introduces a data centric SSL loop that repairs sparse representation regions with frozen diffusion augmentation rather than changing the SSL loss. Second, it replaces pure tail selection with a histogram based mode window that better targets semantically recoverable sparse regions. Third, it provides a broad empirical study across five source regimes, seven downstream tasks, three seeds, competitive long tail SSL baselines, and five targeted ablation groups.

2 Related Work

Self supervised visual representation learning. Modern vision SSL is dominated by contrastive methods such as SimCLR and MoCo [Chen et al., 2020, He et al., 2020], self distillation methods such as DINO [Caron et al., 2021], and masked prediction methods such as MAE [He et al., 2022]. These methods have made unlabeled pretraining routine, but they usually treat the source dataset as fixed and place most of the burden on the objective.

Long tailed and imbalanced SSL. Long tailed data changes both optimization and representation geometry. TS addresses this with a schedule over the contrastive temperature [Kukleva et al., 2023]. SDCLR prunes a self competitor to emphasize hard and tail examples [Jiang et al., 2021]. FASSL introduces frequency aware SSL training [Lin et al., 2023]. COLT injects external OOD data into long tail SSL [Bai et al., 2023]. Our setting is different. BRIDGE does not require class frequencies, does not modify the core SSL loss, and does not require a separate external OOD collection. It instead repairs sparse regions using images generated from the data already being trained on.

OOD scoring and data curation. Data centric AI emphasizes that dataset composition can be as important as model design [Whang et al., 2023]. For our setting, the scoring rule must work without labels, density estimation assumptions, or class frequency estimates. Mean deep nearest neighbor distance satisfies these constraints and has been effective for OOD detection in high dimensional

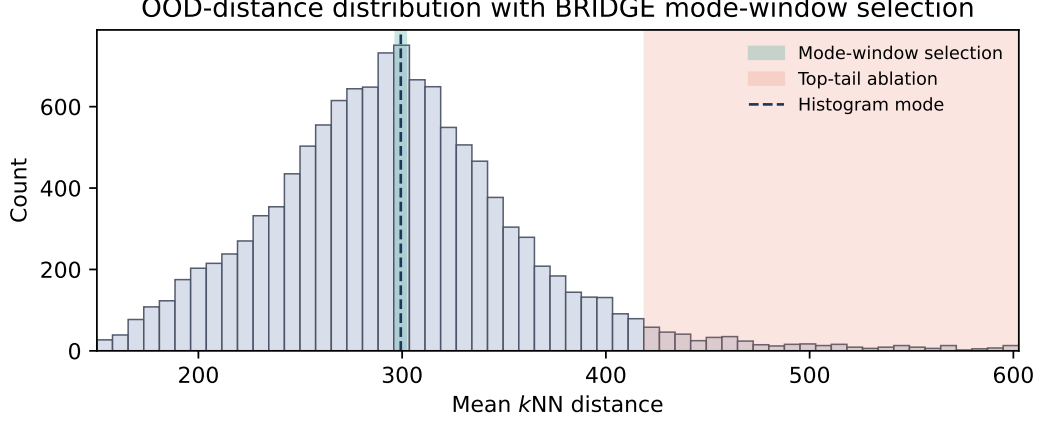


Figure 2: Representative OOD histogram and the selector used in BRIDGE. After clipping the empirical distance distribution to the 1st to 99th percentile range, BRIDGE selects samples closest to the modal bin center. The top tail rule is shown only as an ablation reference.

representation spaces [Sun et al., 2022]. Our contribution is to use this signal not only for detection, but for targeted repair of underrepresented regions.

Diffusion based augmentation. Diffusion models enable high quality image synthesis with strong semantic control [Ho et al., 2020, Rombach et al., 2022]. Prior long tail work has often used generated images in supervised settings. Our use is different. A frozen image to image diffusion model serves as a repair operator inside an SSL loop, and generation is targeted at sparse representation regions instead of globally sampled prompts.

3 Method

3.1 Setting and repair loop

Let $D^{(0)} = \{x_i\}_{i=1}^{N_0}$ denote an unlabeled source dataset and let f_θ be an SSL encoder trained with objective $\mathcal{L}_{ssl}$. BRIDGE alternates between pretraining and dataset repair over C cycles. At cycle c , the current encoder is $f_{\theta^{(c)}}$ and the current source set is $D^{(c)}$.

After training on $D^{(c)}$, we embed every image and compute a non parametric OOD score from mean k NN distance,

$$z_i^{(c)} = f_{\theta^{(c)}}(x_i), \quad d_i^{(c)} = \frac{1}{k} \sum_{j=1}^k \left\| z_i^{(c)} - \pi_j \left(z_i^{(c)} \right) \right\|_2^2, \quad (1)$$

where $\pi_j \left(z_i^{(c)} \right)$ is the j th nearest neighbor of $z_i^{(c)}$ inside the current training set. Large values indicate that an image lies in a sparse part of the learned representation space.

3.2 Mode window OOD selection

The old version of our method selected the largest OOD scores directly. That rule is simple but brittle. The farthest samples are often visually noisy, outlying even relative to the sparse regions that are still semantically useful for repair.

Our current selector therefore works in two steps. First, it clips the empirical OOD distribution to its 1% to 99% quantile range and builds a histogram over the clipped values. Let b^* denote the modal bin and let $m^{(c)}$ denote the center of that bin. Second, given an augmentation budget B , we select the B samples whose OOD scores are closest to $m^{(c)}$,

$$S^{(c)} = \arg \text{top-} B_i - \left| d_i^{(c)} - m^{(c)} \right|. \quad (2)$$

This yields a symmetric selection around the densest sparse region rather than a direct grab of the farthest tail. Figure 2 illustrates the mechanism, and Appendix Figure 3 shows that the OOD shoulder can move substantially across training.

3.3 Diffusion based repair

Given the selected set $S^{(c)}$, we generate G synthetic variants per selected image with a frozen image to image diffusion model g_ϕ . In the main method we use Stable Diffusion 3. The generation ablation replaces it with FLUX or with a no generation re population baseline. The repaired source set is

$$D^{(c+1)} = D^{(c)} \cup \left\{ g_\phi(x, \epsilon_j) \mid x \in S^{(c)}, j = 1, \dots, G \right\}. \quad (3)$$

Because the generator is frozen, the learned component remains the SSL encoder. The repair loop is therefore compatible with any SSL objective that can provide embeddings for OOD scoring.

3.4 Design rationale

The key choice is that we do not try to densify the most isolated samples at all costs. The farthest tail often contains degenerate crops, artifacts, or images whose semantics are too unstable for reliable image to image augmentation. The histogram mode targets a more useful region. It identifies images that are underrepresented relative to the main body of the representation space, yet still belong to a coherent local neighborhood. In the completed selection ablation, this change improves the full seven task average over the older top tail rule for both linear probe and k NN transfer. Uniform sampling remains competitive and even slightly stronger on average in that isolated ablation, so we view the mode window as a principled replacement for extreme tail selection rather than as a universally dominant selector.

4 Experimental Setup

Source datasets. We pretrain on five unlabeled source regimes. Three are long tailed image classification datasets, namely ImageNet-100-LT, CIFAR-10-LT, and CIFAR-100-LT. Two are web scale unlabeled corpora, namely PASS [Asano et al., 2021] and DiffusionDB [Wang et al., 2023]. For PASS and DiffusionDB we use 10k example subsets and override the pretraining split to [1.0, 0.0, 0.0], so all sampled images are used for SSL pretraining.

Downstream evaluation. We evaluate transfer on seven benchmarks: CIFAR-10-LT, CIFAR-100-LT, Stanford Cars [Krause et al., 2013], FGVC Aircraft [Maji et al., 2013], Oxford Flowers [Nilsback and Zisserman, 2008], Oxford-IIIT Pets [Parkhi et al., 2012], and ImageNet-100-LT. The main paper reports linear probe results. Full linear probe and k NN tables are deferred to the appendix.

Training protocol. All reported numbers are mean $\pm$ standard deviation over three seeds. The main backbone is ResNet-50. Unless noted otherwise, BRIDGE uses $k = 100$ for OOD scoring, selects 500 images per cycle, generates 5 images per selected image, and trains for 5 cycles with 100 epochs per cycle. The cycle ablation changes the number of cycles while adjusting the per cycle budget accordingly. The successful DINO ablation uses two local crops. Appendix B gives additional implementation details, runtime summaries, and asset notes.

Compared methods. We compare SimCLR, TS, SDCLR, BRIDGE, and two hybrids that combine BRIDGE with TS or SDCLR. These are meaningful comparisons because BRIDGE changes the data while TS and SDCLR change the optimization dynamics.

Ablation protocol. All ablations are run on ImageNet-100-LT source pretraining with the same downstream suite and three seed evaluation. Each ablation changes one component at a time. The pretraining ablation swaps SimCLR, MoCo, and DINO. The generation ablation compares Stable Diffusion 3, FLUX, and no generation re population. The selection ablation compares the new mode window rule, the old top tail rule that takes the most OOD examples directly, and uniform sampling. The cycle ablation varies the number of repair cycles. The architecture ablation compares ResNet-18, ResNet-50, ViT-S, and ViT-B.

Table 1: Linear probe comparison between balanced pretraining, imbalanced pretraining, and BRIDGE on ImageNet-100-LT. The reported average is over all seven downstream linear-probe tasks, including Cars and Aircraft.

Method	C10	C100	Flowers	Pets	IN100-LT	Avg.
Balanced	65.3 $\pm$ 1.2	37.0 $\pm$ 0.8	22.4 $\pm$ 1.6	31.2 $\pm$ 2.0	39.0 $\pm$ 1.0	31.1 $\pm$ 1.1
Imbalanced	65.4 $\pm$ 4.2	38.6 $\pm$ 1.4	23.7 $\pm$ 1.0	28.4 $\pm$ 1.3	39.6 $\pm$ 0.3	31.2 $\pm$ 1.4
BRIDGE (ours)	70.7 $\pm$ 1.7	43.3 $\pm$ 0.6	26.6 $\pm$ 2.7	31.2 $\pm$ 3.7	42.0 $\pm$ 1.2	34.3 $\pm$ 1.8

Table 2: Average linear-probe transfer over all seven downstream tasks. The full per-task linear and k NN tables are deferred to the appendix.

Source	SimCLR	TS	SDCLR	BRIDGE	BRIDGE+TS	BRIDGE+SDCLR
ImageNet-100-LT	39.8 $\pm$ 1.5	39.6 $\pm$ 1.1	15.1 $\pm$ 0.8	42.3 $\pm$ 1.0	35.7 $\pm$ 1.0	15.2 $\pm$ 0.8
CIFAR-10-LT	29.1 $\pm$ 0.9	26.2 $\pm$ 1.3	13.8 $\pm$ 1.1	32.2 $\pm$ 1.6	28.5 $\pm$ 0.9	13.3 $\pm$ 0.9
CIFAR-100-LT	23.8 $\pm$ 1.4	21.9 $\pm$ 1.2	13.2 $\pm$ 0.7	28.7 $\pm$ 1.8	27.1 $\pm$ 2.6	13.2 $\pm$ 1.0
PASS-10k	36.8 $\pm$ 0.8	37.3 $\pm$ 0.9	14.8 $\pm$ 0.9	40.0 $\pm$ 1.0	36.0 $\pm$ 1.1	16.1 $\pm$ 0.8
DiffusionDB-10k	37.7 $\pm$ 0.9	37.8 $\pm$ 0.8	15.2 $\pm$ 0.9	40.3 $\pm$ 1.2	35.4 $\pm$ 1.0	15.5 $\pm$ 1.1

5 Results

5.1 Balanced and imbalanced baselines

Table 1 compares balanced pretraining, imbalanced pretraining, and BRIDGE on ImageNet-100-LT source data. The difference between balanced and imbalanced SimCLR alone is small in this protocol, which suggests that simply discarding majority data is not a sufficient fix. BRIDGE improves the seven task average linear probe score by 3.05 points over the imbalanced baseline and by 3.14 points over the balanced baseline. The gains are broad rather than isolated. Relative to imbalanced SimCLR, BRIDGE improves CIFAR-10-LT by 5.37 points, CIFAR-100-LT by 4.71 points, Flowers by 2.83 points, and ImageNet-100-LT transfer by 2.35 points.

5.2 Comparison across source datasets

Table 2 first reports the seven task average for each source regime. BRIDGE is the strongest average transfer method in all five regimes that we evaluate. Relative to source matched SimCLR, the average gain is +2.41 points on ImageNet-100-LT, +3.15 on CIFAR-10-LT, +4.89 on CIFAR-100-LT, +3.15 on PASS, and +2.54 on DiffusionDB.

Table 3 then shows the per target linear probe comparison for the five displayed targets, while the Cars and Aircraft columns are deferred to the appendix. The detailed pattern is consistent with the average table, but it also clarifies where each competitor is strongest. BRIDGE is best on the overall average for every source regime. TS or BRIDGE+TS is best on the in domain ImageNet-100-LT target in three of the five source regimes, but that advantage does not translate into the best cross task average. SDCLR and BRIDGE+SDCLR remain substantially weaker throughout this protocol.

5.3 Ablations

We now examine each design choice separately on ImageNet-100-LT source pretraining. Each ablation changes one component of BRIDGE while holding the rest of the pipeline fixed. The main text reports the most informative linear probe tables. The appendix gives the full linear probe and k NN versions.

Table 4 provides a compact overview of the main ablation winners. The detailed tables that follow are the ones we use for interpretation.

SSL objective. Table 5 compares SimCLR, MoCo, and DINO inside the BRIDGE loop. SimCLR is best on every reported metric by a large margin. MoCo is a distant second. DINO remains the weakest option even after the successful final configuration that reduced the number of local crops. In this setting, the repair loop is therefore most effective when paired with a standard contrastive objective rather than momentum contrast or self distillation.

Table 3: Linear-probe transfer for all source datasets. PASS and DiffusionDB use 10k examples with a 100/0/0 pretraining split. For space, Cars and Aircraft are omitted from the displayed columns and are deferred to the appendix. The average is computed over all seven downstream linear-probe tasks.

Source	Method	C10	C100	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	73.9 $\pm$ 0.5	46.8 $\pm$ 0.4	35.9 $\pm$ 7.2	38.6 $\pm$ 1.5	50.4 $\pm$ 0.5	39.8 $\pm$ 1.5
	TS	72.5 $\pm$ 0.4	44.0 $\pm$ 0.5	35.7 $\pm$ 1.8	40.6 $\pm$ 2.2	56.7 $\pm$ 1.2	39.6 $\pm$ 1.1
	SDCLR	41.1 $\pm$ 1.7	18.0 $\pm$ 0.6	10.8 $\pm$ 1.7	10.4 $\pm$ 0.3	21.9 $\pm$ 0.6	15.1 $\pm$ 0.8
	BRIDGE (ours)	75.7 $\pm$ 0.2	48.8 $\pm$ 0.3	36.4 $\pm$ 2.4	42.7 $\pm$ 1.9	53.7 $\pm$ 0.6	42.3 $\pm$ 1.0
	BRIDGE+TS (ours)	73.2 $\pm$ 0.1	45.5 $\pm$ 0.4	26.4 $\pm$ 2.7	34.4 $\pm$ 1.7	47.6 $\pm$ 0.6	35.7 $\pm$ 1.0
	BRIDGE+SDCLR (ours)	41.5 $\pm$ 0.5	17.7 $\pm$ 0.4	12.1 $\pm$ 0.8	10.5 $\pm$ 1.9	20.7 $\pm$ 1.0	15.2 $\pm$ 0.8
CIFAR-10-LT	SimCLR	67.8 $\pm$ 0.2	38.0 $\pm$ 1.0	19.2 $\pm$ 1.6	24.0 $\pm$ 1.3	32.0 $\pm$ 0.7	29.1 $\pm$ 0.9
	TS	68.5 $\pm$ 1.3	36.0 $\pm$ 1.7	13.1 $\pm$ 2.0	19.7 $\pm$ 2.3	31.5 $\pm$ 1.2	26.2 $\pm$ 1.3
	SDCLR	39.6 $\pm$ 1.1	15.6 $\pm$ 0.2	12.0 $\pm$ 1.6	9.1 $\pm$ 1.8	16.5 $\pm$ 1.6	13.8 $\pm$ 1.1
	BRIDGE (ours)	70.7 $\pm$ 2.1	41.5 $\pm$ 0.6	24.0 $\pm$ 3.2	29.4 $\pm$ 2.1	34.4 $\pm$ 1.5	32.2 $\pm$ 1.6
	BRIDGE+TS (ours)	73.1 $\pm$ 1.0	42.0 $\pm$ 0.3	10.2 $\pm$ 1.6	23.2 $\pm$ 1.2	34.2 $\pm$ 1.1	28.5 $\pm$ 0.9
	BRIDGE+SDCLR (ours)	39.1 $\pm$ 0.2	15.3 $\pm$ 1.0	8.9 $\pm$ 2.6	9.5 $\pm$ 1.1	16.8 $\pm$ 0.7	13.3 $\pm$ 0.9
CIFAR-100-LT	SimCLR	59.0 $\pm$ 1.5	31.1 $\pm$ 1.6	12.6 $\pm$ 3.4	20.1 $\pm$ 1.0	28.7 $\pm$ 1.3	23.8 $\pm$ 1.4
	TS	56.3 $\pm$ 0.2	28.0 $\pm$ 0.1	15.5 $\pm$ 2.5	16.2 $\pm$ 3.1	26.2 $\pm$ 1.8	21.9 $\pm$ 1.2
	SDCLR	37.1 $\pm$ 0.8	15.7 $\pm$ 0.2	9.4 $\pm$ 1.0	9.9 $\pm$ 1.7	16.2 $\pm$ 0.3	13.2 $\pm$ 0.7
	BRIDGE (ours)	66.8 $\pm$ 3.1	38.3 $\pm$ 2.2	19.8 $\pm$ 0.4	22.7 $\pm$ 2.3	32.6 $\pm$ 1.4	28.7 $\pm$ 1.8
	BRIDGE+TS (ours)	66.4 $\pm$ 3.3	38.0 $\pm$ 3.7	17.2 $\pm$ 3.6	21.6 $\pm$ 3.3	31.4 $\pm$ 2.3	27.1 $\pm$ 2.6
	BRIDGE+SDCLR (ours)	39.1 $\pm$ 1.9	14.9 $\pm$ 0.9	10.9 $\pm$ 2.5	7.9 $\pm$ 0.7	16.0 $\pm$ 0.3	13.2 $\pm$ 1.0
PASS-10k	SimCLR	72.3 $\pm$ 0.2	45.5 $\pm$ 0.5	33.3 $\pm$ 1.3	33.3 $\pm$ 0.9	44.7 $\pm$ 0.9	36.8 $\pm$ 0.8
	TS	71.5 $\pm$ 0.2	43.5 $\pm$ 0.4	33.1 $\pm$ 2.1	36.4 $\pm$ 1.8	49.4 $\pm$ 0.4	37.3 $\pm$ 0.9
	SDCLR	42.2 $\pm$ 1.1	17.7 $\pm$ 0.8	10.6 $\pm$ 1.3	8.2 $\pm$ 1.6	20.8 $\pm$ 0.4	14.8 $\pm$ 0.9
	BRIDGE (ours)	74.2 $\pm$ 0.2	47.4 $\pm$ 0.7	39.4 $\pm$ 3.4	35.8 $\pm$ 0.1	49.5 $\pm$ 1.0	40.0 $\pm$ 1.0
	BRIDGE+TS (ours)	69.1 $\pm$ 0.4	40.5 $\pm$ 0.8	32.7 $\pm$ 4.1	30.8 $\pm$ 1.0	52.6 $\pm$ 0.5	36.0 $\pm$ 1.1
	BRIDGE+SDCLR (ours)	45.5 $\pm$ 1.0	19.3 $\pm$ 0.2	11.0 $\pm$ 1.6	11.3 $\pm$ 1.2	20.9 $\pm$ 0.9	16.1 $\pm$ 0.8
DiffusionDB-10k	SimCLR	74.4 $\pm$ 0.5	48.6 $\pm$ 0.2	31.2 $\pm$ 2.4	36.1 $\pm$ 1.3	45.3 $\pm$ 0.5	37.7 $\pm$ 0.9
	TS	72.3 $\pm$ 0.3	44.4 $\pm$ 0.3	33.1 $\pm$ 1.1	39.9 $\pm$ 2.7	48.8 $\pm$ 0.2	37.8 $\pm$ 0.8
	SDCLR	42.4 $\pm$ 0.2	18.4 $\pm$ 0.2	13.0 $\pm$ 3.7	8.2 $\pm$ 1.4	21.0 $\pm$ 0.6	15.2 $\pm$ 0.9
	BRIDGE (ours)	76.2 $\pm$ 0.1	49.8 $\pm$ 0.1	36.1 $\pm$ 2.7	38.0 $\pm$ 3.2	48.9 $\pm$ 0.8	40.3 $\pm$ 1.2
	BRIDGE+TS (ours)	68.8 $\pm$ 0.1	40.2 $\pm$ 0.4	31.2 $\pm$ 2.4	34.8 $\pm$ 1.6	51.0 $\pm$ 0.6	35.4 $\pm$ 1.0
	BRIDGE+SDCLR (ours)	42.2 $\pm$ 0.7	17.6 $\pm$ 0.3	11.9 $\pm$ 3.6	11.7 $\pm$ 1.3	21.3 $\pm$ 0.8	15.5 $\pm$ 1.1

Table 4: Summary of the most important ablations on ImageNet-100-LT. Average columns are computed over all seven downstream tasks.

Group	Setting	Avg. Lin.	Avg. kNN	Flowers	Pets	IN100-LT
SSL objective	SimCLR	34.7 $\pm$ 1.2	28.1 $\pm$ 0.6	27.2 $\pm$ 1.5	33.2 $\pm$ 3.3	41.7 $\pm$ 1.3
	MoCo	17.8 $\pm$ 1.9	20.5 $\pm$ 0.8	10.4 $\pm$ 3.2	15.0 $\pm$ 1.8	26.6 $\pm$ 1.9
	DINO	15.2 $\pm$ 1.3	10.4 $\pm$ 0.4	12.6 $\pm$ 4.4	8.7 $\pm$ 1.6	20.6 $\pm$ 1.1
Generation	Stable Diffusion 3	41.4 $\pm$ 0.8	32.7 $\pm$ 0.4	37.4 $\pm$ 2.0	41.2 $\pm$ 0.7	51.8 $\pm$ 1.1
	FLUX	35.0 $\pm$ 1.9	28.8 $\pm$ 1.0	29.4 $\pm$ 4.6	31.5 $\pm$ 2.9	43.7 $\pm$ 1.4
	Re-population	32.6 $\pm$ 1.7	26.5 $\pm$ 0.5	25.7 $\pm$ 1.4	31.7 $\pm$ 3.8	40.6 $\pm$ 0.5
Selection	Mode window	34.6 $\pm$ 1.3	28.1 $\pm$ 0.6	27.9 $\pm$ 3.1	31.6 $\pm$ 3.3	41.7 $\pm$ 1.3
	Top tail	34.3 $\pm$ 1.2	27.3 $\pm$ 0.6	27.7 $\pm$ 1.0	30.5 $\pm$ 1.2	42.2 $\pm$ 1.4
	Uniform	35.5 $\pm$ 1.2	28.4 $\pm$ 0.6	32.9 $\pm$ 0.4	31.7 $\pm$ 3.9	42.0 $\pm$ 0.9

Generation mechanism. Table 6 compares Stable Diffusion 3, FLUX, and a no generation re population baseline. Stable Diffusion 3 is best on every reported metric and improves the seven task average by 8.81 points over re population. FLUX is better than no generation, but it still trails Stable Diffusion 3 substantially. This result indicates that BRIDGE needs semantically faithful generation, not only a mechanism that increases data volume.

Sample selection. Table 7 compares the new mode window selector, the old top tail rule that directly takes the most OOD examples, and uniform sampling. The mode window rule improves over the older top tail rule on the average and on most of the shown targets, which justifies replacing the extreme tail variant. Uniform sampling is still the strongest option in this isolated ablation on average, Flowers, and Pets. Taken together, these results indicate that mode window selection is a better replacement for extreme tail selection while preserving the intended repair behavior.

Cycle count. Table 8 varies the number of repair cycles. Performance improves from 2 to 5 to 10 to 20 cycles on the overall average, with 20 cycles producing the strongest average transfer and the best results on CIFAR-10-LT, CIFAR-100-LT, Pets, and ImageNet-100-LT. Two cycles already recover some benefit, but the best results appear only once the repair loop has time to iterate. We

Table 5: Effect of the SSL pretraining objective inside BRIDGE on ImageNet-100-LT. For space, Cars and Aircraft are omitted from the displayed columns and are deferred to the appendix.

Setting	C10	C100	Flowers	Pets	IN100-LT	Avg.
SimCLR	71.8 $\pm$ 1.2	43.5 $\pm$ 0.5	27.2 $\pm$ 1.5	33.2 $\pm$ 3.3	41.7 $\pm$ 1.3	34.7 $\pm$ 1.2
MoCo	46.6 $\pm$ 3.1	18.4 $\pm$ 2.0	10.4 $\pm$ 3.2	15.0 $\pm$ 1.8	26.6 $\pm$ 1.9	17.8 $\pm$ 1.9
DINO	43.6 $\pm$ 0.7	17.8 $\pm$ 0.6	12.6 $\pm$ 4.4	8.7 $\pm$ 1.6	20.6 $\pm$ 1.1	15.2 $\pm$ 1.3

Table 6: Effect of the augmentation mechanism inside BRIDGE on ImageNet-100-LT. For space, Cars and Aircraft are omitted from the displayed columns and are deferred to the appendix.

Setting	C10	C100	Flowers	Pets	IN100-LT	Avg.
Stable Diffusion 3	75.0 $\pm$ 0.2	48.4 $\pm$ 0.2	37.4 $\pm$ 2.0	41.2 $\pm$ 0.7	51.8 $\pm$ 1.1	41.4 $\pm$ 0.8
FLUX	70.7 $\pm$ 0.4	42.1 $\pm$ 1.4	29.4 $\pm$ 4.6	31.5 $\pm$ 2.9	43.7 $\pm$ 1.4	35.0 $\pm$ 1.9
Re-population	67.0 $\pm$ 2.6	39.4 $\pm$ 2.2	25.7 $\pm$ 1.4	31.7 $\pm$ 3.8	40.6 $\pm$ 0.5	32.6 $\pm$ 1.7

185 nevertheless keep 5 cycles as the default in the broad comparison because it offers the best balance
186 between quality and tractable wall time across all source regimes.

187 **Encoder architecture.** Table 9 compares ResNet-18, ResNet-50, ViT-S, and ViT-B. ResNet-50
188 gives the best average transfer and is strongest on CIFAR-10-LT, CIFAR-100-LT, and Pets. ViT-B
189 is the best model on Flowers and ImageNet-100-LT, but its overall average remains slightly below
190 ResNet-50. This is why the final comparison in the main paper uses ResNet-50 as the default
191 backbone.

Table 7: Effect of sample-selection strategy inside BRIDGE on ImageNet-100-LT. For space, Cars and Aircraft are omitted from the displayed columns and are deferred to the appendix.

Setting	C10	C100	Flowers	Pets	IN100-LT	Avg.
Mode window	71.9 $\pm$ 0.3	43.4 $\pm$ 0.7	27.9 $\pm$ 3.1	31.6 $\pm$ 3.3	41.7 $\pm$ 1.3	34.6 $\pm$ 1.3
Top tail	70.8 $\pm$ 0.5	43.1 $\pm$ 1.0	27.7 $\pm$ 1.0	30.5 $\pm$ 1.2	42.2 $\pm$ 1.4	34.3 $\pm$ 1.2
Uniform	71.5 $\pm$ 0.2	43.8 $\pm$ 0.7	32.9 $\pm$ 0.4	31.7 $\pm$ 3.9	42.0 $\pm$ 0.9	35.5 $\pm$ 1.2

Table 8: Effect of cycle count inside BRIDGE on ImageNet-100-LT. For space, Cars and Aircraft are omitted from the displayed columns and are deferred to the appendix.

Setting	C10	C100	Flowers	Pets	IN100-LT	Avg.
2 cycles	67.9 $\pm$ 0.6	40.7 $\pm$ 0.6	30.5 $\pm$ 3.2	27.4 $\pm$ 1.7	40.1 $\pm$ 0.4	33.1 $\pm$ 1.4
5 cycles	70.7 $\pm$ 1.7	43.3 $\pm$ 0.6	26.6 $\pm$ 2.7	31.2 $\pm$ 3.7	42.0 $\pm$ 1.2	34.3 $\pm$ 1.8
10 cycles	70.5 $\pm$ 1.2	43.8 $\pm$ 0.6	24.8 $\pm$ 5.2	31.3 $\pm$ 1.7	41.9 $\pm$ 0.6	34.1 $\pm$ 1.7
20 cycles	71.6 $\pm$ 1.1	44.6 $\pm$ 0.9	27.0 $\pm$ 2.1	33.3 $\pm$ 4.4	42.1 $\pm$ 1.5	35.1 $\pm$ 1.6

Table 9: Effect of encoder architecture inside BRIDGE on ImageNet-100-LT. For space, Cars and Aircraft are omitted from the displayed columns and are deferred to the appendix.

Setting	C10	C100	Flowers	Pets	IN100-LT	Avg.
ResNet-18	67.7 $\pm$ 0.8	40.4 $\pm$ 0.5	26.1 $\pm$ 2.4	29.5 $\pm$ 2.7	41.0 $\pm$ 1.2	32.9 $\pm$ 1.2
ResNet-50	71.7 $\pm$ 1.0	44.0 $\pm$ 1.1	28.8 $\pm$ 1.7	32.0 $\pm$ 1.6	41.7 $\pm$ 1.3	34.9 $\pm$ 1.1
ViT-S	62.7 $\pm$ 0.4	37.6 $\pm$ 0.6	35.1 $\pm$ 4.2	30.2 $\pm$ 1.6	42.5 $\pm$ 1.2	32.7 $\pm$ 1.3
ViT-B	66.0 $\pm$ 2.3	40.6 $\pm$ 2.3	38.8 $\pm$ 2.6	31.3 $\pm$ 2.1	44.5 $\pm$ 1.7	34.8 $\pm$ 1.8

6 Limitations

BRIDGE is more expensive than fixed data SSL because every cycle requires feature extraction, OOD scoring, and image generation. The successful experiments reported here use a specific compute regime based on A100 nodes and 24 hour wall time limits, and Appendix C shows that total project compute is substantial. The PASS and DiffusionDB experiments are limited to 10k subsets for tractability. The main paper also centers the final comparison on ResNet-50, even though the appendix includes broader architecture ablations. Finally, the submission does not benchmark newer methods that appeared after the experimental campaign was completed.

7 Conclusion

The experiments support a simple conclusion. Long tail SSL is not only an optimization problem. It is also a data geometry problem. By repeatedly identifying sparse representation regions and repairing them with frozen diffusion augmentation, BRIDGE improves average transfer across conventional long tailed sources and across two web scale unlabeled sources. The new mode window selector matters because it replaces an overly literal extreme tail rule with a more stable target region. Within the completed experimental protocol, BRIDGE is the strongest average transfer method across all five source regimes that we study.

References

- Yuki Asano, Christian Rupprecht, Andrew Zisserman, and Andrea Vedaldi. PASS: An ImageNet replacement for self-supervised pretraining without humans. In Joaquin Vanschoren and Serena Yeung, editors, *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, volume 1, 2021. URL https://datasets-benchmarks-proceedings.neurips.cc/paper_files/paper/2021/file/0777d5c17d4066b82ab86dff8a46af6f-Paper-round1.pdf.
- Jianhong Bai, Zuozhu Liu, Hualiang Wang, Jin Hao, Yan Feng, Huanpeng Chu, and Haoji Hu. On the effectiveness of out-of-distribution data in self-supervised long-tail learning. In *The Eleventh International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=v8JIQdiN9Sh>. Published as a conference paper at ICLR 2023.
- Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9650–9660. IEEE, October 2021. doi: 10.1109/ICCV48922.2021.00951. URL https://openaccess.thecvf.com/content/ICCV2021/html/Caron_Emerging_Properties_in_Self-Supervised_Vision_Transformers_ICCV_2021_paper.html.
- Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In Hal Daumé III and Aarti Singh, editors, *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 1597–1607. PMLR, 13–18 Jul 2020. URL <https://proceedings.mlr.press/v119/chen20j.html>.
- Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. Momentum contrast for unsupervised visual representation learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9729–9738. IEEE, June 2020. doi: 10.1109/CVPR42600.2020.00975. URL https://openaccess.thecvf.com/content_CVPR_2020/html/He_Momentum_Contrast_for_Unsupervised_Visual_Representation_Learning_CVPR_2020_paper.html.
- Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16000–16009. IEEE, June 2022. URL https://openaccess.thecvf.com/content/CVPR2022/html/He_Masked_Autoencoders_Are_Scalable_Vision_Learners_CVPR_2022_paper.html.

- Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In Hugo Larochelle, Marc’Aurelio Ranzato, Raia Hadsell, Maria-Florina Balcan, and Hsuan-Tien Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/file/4c5bcfec8584af0d967f1ab10179ca4b-Paper.pdf.
- Ziyu Jiang, Tianlong Chen, Bobak J. Mortazavi, and Zhangyang Wang. Self-damaging contrastive learning. In Marina Meila and Tong Zhang, editors, *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 4927–4939. PMLR, 18–24 Jul 2021. URL <https://proceedings.mlr.press/v139/jiang21a.html>.
- Jonathan Krause, Michael Stark, Jia Deng, and Li Fei-Fei. 3d object representations for fine-grained categorization. In *2013 IEEE International Conference on Computer Vision Workshops*, pages 554–561. IEEE, 2013. doi: 10.1109/ICCVW.2013.77. URL <https://doi.org/10.1109/ICCVW.2013.77>.
- Anna Kukleva, Moritz Böhle, Bernt Schiele, Hilde Kuehne, and Christian Rupprecht. Temperature schedules for self-supervised contrastive methods on long-tail data. In *The Eleventh International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=ejHUr4nfHhD>. Published as a conference paper at ICLR 2023.
- Ci-Siang Lin, Min-Hung Chen, and Yu-Chiang Frank Wang. Frequency-aware self-supervised long-tailed learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*, pages 963–972. IEEE, October 2023. doi: 10.1109/ICCVW60793.2023.00103. URL https://openaccess.thecvf.com/content/ICCV2023W/LIMIT/html/Lin_Frequency-Aware_Self-Supervised_Long-Tailed_Learning_ICCVW_2023_paper.html.
- Subhansu Maji, Esa Rahtu, Juho Kannala, Matthew B. Blaschko, and Andrea Vedaldi. Fine-grained visual classification of aircraft, 2013. URL <https://arxiv.org/abs/1306.5151>.
- Maria-Elena Nilsback and Andrew Zisserman. Automated flower classification over a large number of classes. In *2008 Sixth Indian Conference on Computer Vision, Graphics & Image Processing*, pages 722–729. IEEE, 2008. doi: 10.1109/ICVGIP.2008.47. URL <https://doi.org/10.1109/ICVGIP.2008.47>.
- Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khaidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jégou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. URL <https://openreview.net/forum?id=a68SUt6zFt>. Published 05/2024.
- Omkar M. Parkhi, Andrea Vedaldi, Andrew Zisserman, and C. V. Jawahar. Cats and dogs. In *2012 IEEE Conference on Computer Vision and Pattern Recognition*, pages 3498–3505. IEEE, 2012. doi: 10.1109/CVPR.2012.6248092. URL <https://doi.org/10.1109/CVPR.2012.6248092>.
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10684–10695. IEEE, June 2022. URL https://openaccess.thecvf.com/content/CVPR2022/html/Rombach_High-Resolution_Image_Synthesis_With_Latent_Diffusion_Models_CVPR_2022_paper.html.
- Yiyou Sun, Yifei Ming, Xiaojin Zhu, and Yixuan Li. Out-of-distribution detection with deep nearest neighbors. In Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvári, Gang Niu, and Sivan Sabato, editors, *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 20827–20840. PMLR, 17–23 Jul 2022. URL <https://proceedings.mlr.press/v162/sun22d.html>.

292 Zijie J. Wang, Evan Montoya, David Munechika, Haoyang Yang, Benjamin Hoover, and
 293 Duen Horng Chau. DiffusionDB: A large-scale prompt gallery dataset for text-to-image generative models. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 893–911, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.51. URL <https://aclanthology.org/2023.acl-long.51/>.
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 295
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 297
 298
 299 Steven Euijong Whang, Yuji Roh, Hwanjun Song, and Jae-Gil Lee. Data collection and quality challenges in deep learning: A data-centric AI perspective. *The VLDB Journal*, 32(4): 791–813, 2023. doi: 10.1007/s00778-022-00775-9. URL <https://doi.org/10.1007/s00778-022-00775-9>.
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303 A Additional Diagnostics

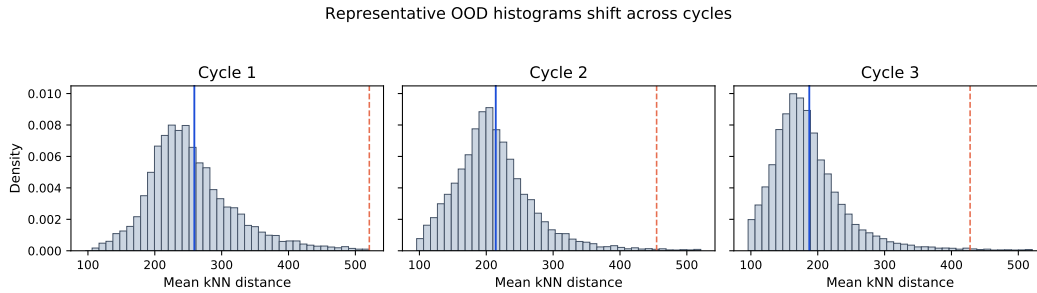


Figure 3: Representative OOD histograms from a successful ImageNet-100-LT TS run. The sparse shoulder moves across cycles, which motivates selecting around a local modal band rather than always taking the farthest tail.

304 B Implementation and Reproducibility

305 **Core training configuration.** The completed main runs use ResNet-50 encoders. Most baseline, BRIDGE, and source regime comparisons use 5 cycles with 100 epochs per cycle. The default BRIDGE configuration uses mean k NN distance with $k = 100$, selects 500 samples per cycle, and generates 5 images per selected sample. The default optimizer is AdamW with weight decay 10^{-4} , and method specific learning rates and temperatures follow the completed job configurations.
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310 **OOD scoring and selection.** Features are extracted without feature normalization and nearest neighbor distances are computed with FAISS using squared ℓ_2 distance. The mode window selector clips the empirical OOD distribution to the 1st to 99th percentile range, builds a histogram with the code level auto bin heuristic, and selects the examples closest to the modal bin center.
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314 **Generators.** The main method uses Stable Diffusion 3 in image to image mode. The generation ablation replaces it with FLUX or with a no generation re population baseline. In the completed PASS and DiffusionDB runs we increase the SD3 batch size to 12 to fit the 24 hour Slurm limit.
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317 **Source data specifics.** PASS and DiffusionDB use `train[:10000]` with pretraining split override `[1.0, 0.0, 0.0]`. This keeps the source data budget aligned across methods while avoiding unnecessary pretraining validation overhead. The long tailed ImageNet-100, CIFAR-10, and CIFAR-100 regimes follow the completed job configurations used in the final sweep.
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321 **Seeds and reporting.** All main claims are supported by three seed averages. Tables report mean $\pm$ standard deviation over those seeds. The exact table values are generated from the completed job logs by the local export scripts in `scripts/export_paper_results.py` and `scripts/generate_neurips_paper_tables.py`.
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Table 10: Full linear-probe baseline comparison on ImageNet-100-LT pretraining.

Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Balanced	65.3 $\pm$ 1.2	37.0 $\pm$ 0.8	8.0 $\pm$ 0.5	15.1 $\pm$ 0.7	22.4 $\pm$ 1.6	31.2 $\pm$ 2.0	39.0 $\pm$ 1.0	31.1 $\pm$ 1.1
Imbalanced	65.4 $\pm$ 4.2	38.6 $\pm$ 1.4	7.6 $\pm$ 1.1	15.3 $\pm$ 0.9	23.7 $\pm$ 1.0	28.4 $\pm$ 1.3	39.6 $\pm$ 0.3	31.2 $\pm$ 1.4
BRIDGE (ours)	70.7 $\pm$ 1.7	43.3 $\pm$ 0.6	9.9 $\pm$ 1.5	16.4 $\pm$ 0.9	26.6 $\pm$ 2.7	31.2 $\pm$ 3.7	42.0 $\pm$ 1.2	34.3 $\pm$ 1.8

Table 11: Full kNN baseline comparison on ImageNet-100-LT pretraining.

Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Balanced	63.6 $\pm$ 0.7	32.8 $\pm$ 0.8	4.4 $\pm$ 0.4	9.0 $\pm$ 0.4	22.3 $\pm$ 0.1	17.2 $\pm$ 0.1	34.0 $\pm$ 1.1	26.2 $\pm$ 0.5
Imbalanced	63.9 $\pm$ 0.2	33.2 $\pm$ 0.3	4.5 $\pm$ 0.6	9.0 $\pm$ 0.3	22.5 $\pm$ 0.3	16.8 $\pm$ 0.2	34.3 $\pm$ 0.3	26.3 $\pm$ 0.3
BRIDGE (ours)	66.0 $\pm$ 0.9	35.0 $\pm$ 0.9	5.0 $\pm$ 0.4	9.8 $\pm$ 0.2	24.8 $\pm$ 0.4	18.0 $\pm$ 0.5	35.6 $\pm$ 1.2	27.7 $\pm$ 0.6

C Compute Resources and Runtime

Most completed experiments use one node with two NVIDIA A100 80GB GPUs and a 24 hour wall time limit. The successful DINO ablation uses four A100 80GB GPUs after reducing the number of local crops from six to two. Over the successful experiments reported in the paper, the aggregate training time is approximately 3022 wall clock hours, or about 6142 GPU hours. This excludes failed runs, timeout runs, cache migration jobs, and other debugging or pilot jobs, so the full project required more compute than the successful totals alone.

Representative mean wall clock times over three seeds are as follows. On ImageNet-100-LT source pretraining, SimCLR takes 16.9 hours and BRIDGE takes 21.8 hours. On PASS, SimCLR takes 10.2 hours and BRIDGE takes 14.9 hours. On DiffusionDB, SimCLR takes 11.6 hours and BRIDGE takes 16.4 hours. The successful DINO ablation takes 16.3 hours on four GPUs in its final configuration.

D Assets and Terms of Use

We use existing datasets and pretrained generators under their original release terms.

Source datasets. ImageNet-100-LT is derived from ImageNet, whose official access terms restrict the images to non commercial research and educational use. PASS is released under Creative Commons Attribution 4.0. DiffusionDB is released under CC0-1.0. CIFAR-10, CIFAR-100, Stanford Cars, FGVC Aircraft, Oxford Flowers, and Oxford-IIIT Pets are used from their official academic benchmark releases. When an official page provides download terms instead of a standard software style license, we follow those original terms.

Generators. Stable Diffusion 3 Medium is used under the Stability AI Community License. FLUX.1-dev is used only in the ablation and is subject to the FLUX.1-dev Non-Commercial License. We do not redistribute model weights, third party datasets, or generated image corpora as part of this submission.

E Broader Impact

A method that improves representation quality under biased source data can reduce the need for manual relabeling and can make SSL more useful for domains where balanced data is difficult to collect. That is a positive outcome. At the same time, our method relies on large image generators and on web sourced datasets. These carry familiar risks related to license compliance, privacy, and unintended memorization or harmful content. Our experiments stay within existing dataset and model release terms and do not release new generators or scraped image collections. Any future deployment would still require a task specific review of fairness, privacy, and content safety.

Table 12: Full linear-probe transfer results across all source datasets and comparison methods.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	73.9 $\pm$ 0.5	46.8 $\pm$ 0.4	13.7 $\pm$ 0.3	19.6 $\pm$ 0.2	35.9 $\pm$ 7.2	38.6 $\pm$ 1.5	50.4 $\pm$ 0.5	39.8 $\pm$ 1.5
	TS	72.5 $\pm$ 0.4	44.0 $\pm$ 0.5	12.5 $\pm$ 1.2	15.0 $\pm$ 0.1	35.7 $\pm$ 1.8	40.6 $\pm$ 2.2	56.7 $\pm$ 1.2	39.6 $\pm$ 1.1
	SDCLR	41.1 $\pm$ 1.7	18.0 $\pm$ 0.6	1.4 $\pm$ 0.3	2.3 $\pm$ 0.5	10.8 $\pm$ 1.7	10.4 $\pm$ 0.3	21.9 $\pm$ 0.6	15.1 $\pm$ 0.8
	BRIDGE (ours)	75.7 $\pm$ 0.2	48.8 $\pm$ 0.3	16.1 $\pm$ 1.0	22.4 $\pm$ 0.3	36.4 $\pm$ 2.4	42.7 $\pm$ 1.9	53.7 $\pm$ 0.6	42.3 $\pm$ 1.0
	BRIDGE+TS (ours)	73.2 $\pm$ 0.1	45.5 $\pm$ 0.4	9.5 $\pm$ 1.2	13.0 $\pm$ 0.7	26.4 $\pm$ 2.7	34.4 $\pm$ 1.7	47.6 $\pm$ 0.6	35.7 $\pm$ 1.0
	BRIDGE+SDCLR (ours)	41.5 $\pm$ 0.5	17.7 $\pm$ 0.4	1.3 $\pm$ 0.3	2.4 $\pm$ 0.7	12.1 $\pm$ 0.8	10.5 $\pm$ 1.9	20.7 $\pm$ 1.0	15.2 $\pm$ 0.8
CIFAR-10-LT	SimCLR	67.8 $\pm$ 0.2	38.0 $\pm$ 1.0	6.5 $\pm$ 0.4	16.0 $\pm$ 0.9	19.2 $\pm$ 1.6	24.0 $\pm$ 1.3	32.0 $\pm$ 0.7	29.1 $\pm$ 0.9
	TS	68.5 $\pm$ 1.3	36.0 $\pm$ 1.7	4.3 $\pm$ 0.2	10.3 $\pm$ 0.1	13.1 $\pm$ 2.0	19.7 $\pm$ 2.3	31.5 $\pm$ 1.2	26.2 $\pm$ 1.3
	SDCLR	39.6 $\pm$ 1.1	15.6 $\pm$ 0.2	1.1 $\pm$ 0.1	2.4 $\pm$ 0.9	12.0 $\pm$ 1.6	9.1 $\pm$ 1.8	16.5 $\pm$ 1.6	13.8 $\pm$ 1.1
	BRIDGE (ours)	70.7 $\pm$ 2.1	41.5 $\pm$ 0.6	8.6 $\pm$ 0.4	17.0 $\pm$ 0.9	24.0 $\pm$ 3.2	29.4 $\pm$ 2.1	34.4 $\pm$ 1.5	32.2 $\pm$ 1.6
	BRIDGE+TS (ours)	73.1 $\pm$ 1.0	42.0 $\pm$ 0.3	4.4 $\pm$ 0.8	12.4 $\pm$ 0.5	10.2 $\pm$ 1.6	23.2 $\pm$ 1.2	34.2 $\pm$ 1.1	28.5 $\pm$ 0.9
	BRIDGE+SDCLR (ours)	39.1 $\pm$ 0.2	15.3 $\pm$ 1.0	1.1 $\pm$ 0.3	2.1 $\pm$ 0.6	8.9 $\pm$ 2.6	9.5 $\pm$ 1.1	16.8 $\pm$ 0.7	13.3 $\pm$ 0.9
CIFAR-100-LT	SimCLR	59.0 $\pm$ 1.5	31.1 $\pm$ 1.6	4.1 $\pm$ 0.3	11.2 $\pm$ 1.0	12.6 $\pm$ 3.4	20.1 $\pm$ 1.0	28.7 $\pm$ 1.3	23.8 $\pm$ 1.4
	TS	56.3 $\pm$ 0.2	28.0 $\pm$ 0.1	2.8 $\pm$ 0.6	8.5 $\pm$ 0.2	15.5 $\pm$ 2.5	16.2 $\pm$ 3.1	26.2 $\pm$ 1.8	21.9 $\pm$ 1.2
	SDCLR	37.1 $\pm$ 0.8	15.7 $\pm$ 0.2	1.9 $\pm$ 0.1	2.4 $\pm$ 0.7	9.4 $\pm$ 1.0	9.9 $\pm$ 1.7	16.2 $\pm$ 0.3	13.2 $\pm$ 0.7
	BRIDGE (ours)	66.8 $\pm$ 3.1	38.3 $\pm$ 2.2	6.2 $\pm$ 1.8	14.6 $\pm$ 1.6	19.8 $\pm$ 0.4	22.7 $\pm$ 2.3	32.6 $\pm$ 1.4	28.7 $\pm$ 1.8
	BRIDGE+TS (ours)	66.4 $\pm$ 3.3	38.0 $\pm$ 3.7	4.2 $\pm$ 1.0	10.9 $\pm$ 0.7	17.2 $\pm$ 3.6	21.6 $\pm$ 3.3	31.4 $\pm$ 2.3	27.1 $\pm$ 2.6
	BRIDGE+SDCLR (ours)	39.1 $\pm$ 1.9	14.9 $\pm$ 0.9	0.9 $\pm$ 0.2	3.0 $\pm$ 0.4	10.9 $\pm$ 2.5	7.9 $\pm$ 0.7	16.0 $\pm$ 0.3	13.2 $\pm$ 1.0
PASS-10k	SimCLR	72.3 $\pm$ 0.2	45.5 $\pm$ 0.5	11.0 $\pm$ 0.2	17.5 $\pm$ 1.4	33.3 $\pm$ 1.3	33.3 $\pm$ 0.9	44.7 $\pm$ 0.9	36.8 $\pm$ 0.8
	TS	71.5 $\pm$ 0.2	43.5 $\pm$ 0.4	10.2 $\pm$ 1.2	17.2 $\pm$ 0.2	33.1 $\pm$ 2.1	36.4 $\pm$ 1.8	49.4 $\pm$ 0.4	37.3 $\pm$ 0.9
	SDCLR	42.2 $\pm$ 1.1	17.7 $\pm$ 0.8	1.8 $\pm$ 0.3	2.2 $\pm$ 0.8	10.6 $\pm$ 1.3	8.2 $\pm$ 1.6	20.8 $\pm$ 0.4	14.8 $\pm$ 0.9
	BRIDGE (ours)	74.2 $\pm$ 0.2	47.4 $\pm$ 0.7	13.0 $\pm$ 1.2	20.4 $\pm$ 0.4	39.4 $\pm$ 3.4	35.8 $\pm$ 0.1	49.5 $\pm$ 1.0	40.0 $\pm$ 1.0
	BRIDGE+TS (ours)	69.1 $\pm$ 0.4	40.5 $\pm$ 0.8	12.1 $\pm$ 0.5	14.3 $\pm$ 0.3	32.7 $\pm$ 4.1	30.8 $\pm$ 1.0	52.6 $\pm$ 0.5	36.0 $\pm$ 1.1
	BRIDGE+SDCLR (ours)	45.5 $\pm$ 1.0	19.3 $\pm$ 0.2	1.5 $\pm$ 0.7	3.3 $\pm$ 0.2	11.0 $\pm$ 1.6	11.3 $\pm$ 1.2	20.9 $\pm$ 0.9	16.1 $\pm$ 0.8
DiffusionDB-10k	SimCLR	74.4 $\pm$ 0.5	48.6 $\pm$ 0.2	10.9 $\pm$ 0.7	17.6 $\pm$ 0.8	31.2 $\pm$ 2.4	36.1 $\pm$ 1.3	45.3 $\pm$ 0.5	37.7 $\pm$ 0.9
	TS	72.3 $\pm$ 0.3	44.4 $\pm$ 0.3	10.7 $\pm$ 0.6	15.4 $\pm$ 0.6	33.1 $\pm$ 1.1	39.9 $\pm$ 2.7	48.8 $\pm$ 0.2	37.8 $\pm$ 0.8
	SDCLR	42.4 $\pm$ 0.2	18.4 $\pm$ 0.2	1.1 $\pm$ 0.2	2.4 $\pm$ 0.1	13.0 $\pm$ 3.7	8.2 $\pm$ 1.4	21.0 $\pm$ 0.6	15.2 $\pm$ 0.9
	BRIDGE (ours)	76.2 $\pm$ 0.1	49.8 $\pm$ 0.1	12.4 $\pm$ 1.1	20.4 $\pm$ 0.2	36.1 $\pm$ 2.7	38.0 $\pm$ 3.2	48.9 $\pm$ 0.8	40.3 $\pm$ 1.2
	BRIDGE+TS (ours)	68.8 $\pm$ 0.1	40.2 $\pm$ 0.4	9.2 $\pm$ 1.1	12.8 $\pm$ 0.5	31.2 $\pm$ 2.4	34.8 $\pm$ 1.6	51.0 $\pm$ 0.6	35.4 $\pm$ 1.0
	BRIDGE+SDCLR (ours)	42.2 $\pm$ 0.7	17.6 $\pm$ 0.3	1.1 $\pm$ 0.6	2.4 $\pm$ 0.1	11.9 $\pm$ 3.6	11.7 $\pm$ 1.3	21.3 $\pm$ 0.8	15.5 $\pm$ 1.1

Table 13: Full kNN transfer results across all source datasets and comparison methods.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	68.6 $\pm$ 0.4	37.3 $\pm$ 0.2	6.0 $\pm$ 0.7	11.2 $\pm$ 0.4	32.2 $\pm$ 0.7	23.6 $\pm$ 0.3	43.9 $\pm$ 0.9	31.8 $\pm$ 0.5
	TS	68.4 $\pm$ 0.3	36.5 $\pm$ 0.1	4.7 $\pm$ 0.2	8.3 $\pm$ 0.5	26.6 $\pm$ 1.0	23.6 $\pm$ 1.7	52.2 $\pm$ 0.5	31.5 $\pm$ 0.6
	SDCLR	32.5 $\pm$ 0.4	10.9 $\pm$ 0.2	1.0 $\pm$ 0.3	2.1 $\pm$ 0.1	10.7 $\pm$ 0.7	4.7 $\pm$ 0.2	11.2 $\pm$ 0.2	10.4 $\pm$ 0.3
	BRIDGE (ours)	68.9 $\pm$ 0.5	37.8 $\pm$ 0.1	6.9 $\pm$ 0.1	12.1 $\pm$ 0.2	36.0 $\pm$ 0.1	25.3 $\pm$ 0.9	48.2 $\pm$ 0.4	33.6 $\pm$ 0.3
	BRIDGE+TS (ours)	65.2 $\pm$ 0.2	33.7 $\pm$ 0.5	2.9 $\pm$ 0.4	6.6 $\pm$ 0.2	18.6 $\pm$ 0.4	15.8 $\pm$ 1.0	40.5 $\pm$ 0.8	26.2 $\pm$ 0.5
	BRIDGE+SDCLR (ours)	33.8 $\pm$ 0.4	11.5 $\pm$ 0.5	1.3 $\pm$ 0.4	2.0 $\pm$ 0.1	10.2 $\pm$ 0.7	4.9 $\pm$ 0.2	12.6 $\pm$ 0.6	10.9 $\pm$ 0.4
CIFAR-10-LT	SimCLR	66.9 $\pm$ 0.4	33.4 $\pm$ 0.3	3.7 $\pm$ 0.4	9.8 $\pm$ 0.7	16.5 $\pm$ 0.8	13.9 $\pm$ 0.8	24.0 $\pm$ 0.8	24.0 $\pm$ 0.6
	TS	66.3 $\pm$ 0.6	27.8 $\pm$ 0.4	2.4 $\pm$ 0.6	6.4 $\pm$ 0.2	10.5 $\pm$ 0.7	11.1 $\pm$ 0.4	22.0 $\pm$ 0.9	20.9 $\pm$ 0.5
	SDCLR	30.1 $\pm$ 0.3	10.0 $\pm$ 0.3	1.7 $\pm$ 0.3	2.2 $\pm$ 0.2	10.1 $\pm$ 0.7	4.4 $\pm$ 0.4	9.8 $\pm$ 0.3	9.7 $\pm$ 0.4
	BRIDGE (ours)	69.0 $\pm$ 1.2	35.2 $\pm$ 0.9	4.2 $\pm$ 0.6	10.4 $\pm$ 0.5	17.0 $\pm$ 0.6	15.2 $\pm$ 0.8	25.2 $\pm$ 0.5	25.2 $\pm$ 0.7
	BRIDGE+TS (ours)	67.7 $\pm$ 0.8	28.0 $\pm$ 0.8	2.1 $\pm$ 0.2	5.1 $\pm$ 0.7	8.1 $\pm$ 1.0	9.1 $\pm$ 0.8	21.8 $\pm$ 0.5	20.3 $\pm$ 0.7
	BRIDGE+SDCLR (ours)	31.3 $\pm$ 1.2	10.4 $\pm$ 0.5	1.0 $\pm$ 0.1	1.8 $\pm$ 0.3	9.6 $\pm$ 1.3	4.5 $\pm$ 0.1	10.1 $\pm$ 0.4	9.8 $\pm$ 0.6
CIFAR-100-LT	SimCLR	59.3 $\pm$ 0.8	30.1 $\pm$ 0.9	3.0 $\pm$ 0.3	7.4 $\pm$ 0.4	13.7 $\pm$ 0.2	10.8 $\pm$ 0.4	21.8 $\pm$ 1.1	20.9 $\pm$ 0.6
	TS	53.1 $\pm$ 0.6	24.6 $\pm$ 0.4	2.4 $\pm$ 0.4	5.5 $\pm$ 0.3	9.3 $\pm$ 0.4	8.3 $\pm$ 0.2	18.7 $\pm$ 1.0	17.4 $\pm$ 0.5
	SDCLR	25.7 $\pm$ 1.2	8.5 $\pm$ 0.6	0.9 $\pm$ 0.4	1.9 $\pm$ 0.2	7.3 $\pm$ 0.3	3.9 $\pm$ 0.3	8.3 $\pm$ 1.1	8.1 $\pm$ 0.6
	BRIDGE (ours)	62.5 $\pm$ 0.8	32.5 $\pm$ 0.3	3.7 $\pm$ 0.3	8.7 $\pm$ 0.5	14.1 $\pm$ 0.9	12.1 $\pm$ 0.7	23.0 $\pm$ 1.2	22.4 $\pm$ 0.7
	BRIDGE+TS (ours)	56.3 $\pm$ 1.2	27.8 $\pm$ 0.5	1.9 $\pm$ 0.0	4.8 $\pm$ 0.6	8.0 $\pm$ 1.0	9.0 $\pm$ 0.6	20.8 $\pm$ 0.8	18.4 $\pm$ 0.7
	BRIDGE+SDCLR (ours)	29.7 $\pm$ 2.2	9.8 $\pm$ 1.5	1.4 $\pm$ 0.4	2.3 $\pm$ 0.2	9.2 $\pm$ 1.1	4.8 $\pm$ 0.1	8.9 $\pm$ 0.6	9.4 $\pm$ 0.9
PASS-10k	SimCLR	67.2 $\pm$ 0.2	37.0 $\pm$ 0.3	5.4 $\pm$ 0.3	10.3 $\pm$ 0.5	30.4 $\pm$ 0.0	16.7 $\pm$ 0.5	38.3 $\pm$ 0.3	29.3 $\pm$ 0.3
	TS	65.2 $\pm$ 0.7	34.6 $\pm$ 0.3	4.4 $\pm$ 0.2	7.9 $\pm$ 0.2	24.5 $\pm$ 0.1	14.4 $\pm$ 0.4	39.3 $\pm$ 0.4	27.2 $\pm$ 0.3
	SDCLR	32.4 $\pm$ 1.5	11.5 $\pm$ 0.8	1.3 $\pm$ 0.2	2.0 $\pm$ 0.1	8.6 $\pm$ 0.7	5.2 $\pm$ 0.2	11.1 $\pm$ 0.8	10.3 $\pm$ 0.6
	BRIDGE (ours)	67.8 $\pm$ 0.6	37.1 $\pm$ 0.5	5.2 $\pm$ 0.4	10.8 $\pm$ 0.3	33.7 $\pm$ 0.6	17.9 $\pm$ 0.3	41.9 $\pm$ 0.4	30.6 $\pm$ 0.5
	BRIDGE+TS (ours)	59.0 $\pm$ 0.4	27.8 $\pm$ 0.5	3.3 $\pm$ 0.4	6.6 $\pm$ 0.0	23.3 $\pm$ 0.4	13.9 $\pm$ 0.3	39.2 $\pm$ 0.3	24.7 $\pm$ 0.3
	BRIDGE+SDCLR (ours)	37.9 $\pm$ 0.7	13.6 $\pm$ 0.0	1.2 $\pm$ 0.2	2.7 $\pm$ 0.1	8.9 $\pm$ 1.3	6.3 $\pm$ 0.1	13.9 $\pm$ 1.1	12.1 $\pm$ 0.5
DiffusionDB-10k	SimCLR	67.3 $\pm$ 0.1	36.2 $\pm$ 0.4	4.3 $\pm$ 0.2	8.8 $\pm$ 0.1	26.8 $\pm$ 0.4	17.3 $\pm$ 0.1	36.7 $\pm$ 0.8	28.2 $\pm$ 0.3
	TS	64.7 $\pm$ 0.4	33.6 $\pm$ 0.3	3.1 $\pm$ 0.5	7.3 $\pm$ 0.3	23.2 $\pm$ 0.4	17.0 $\pm$ 0.0	37.8 $\pm$ 0.6	26.7 $\pm$ 0.4
	SDCLR	32.6 $\pm$ 1.2	10.0 $\pm$ 0.3	1.3 $\pm$ 0.1	2.2 $\pm$ 0.2	6.1 $\pm$ 0.7	4.8 $\pm$ 0.4	10.6 $\pm$ 0.5	9.7 $\pm$ 0.5
	BRIDGE (ours)	68.1 $\pm$ 0.3	37.0 $\pm$ 0.2	5.4 $\pm$ 0.5	9.5 $\pm$ 0.1	30.4 $\pm$ 0.4	18.5 $\pm$ 0.1	39.9 $\pm$ 0.6	29.8 $\pm$ 0.3
	BRIDGE+TS (ours)	57.2 $\pm$ 0.7	25.6 $\pm$ 1.0	2.8 $\pm$ 0.3	6.8 $\pm$ 0.1	22.3 $\pm$ 0.5	15.4 $\pm$ 0.6	37.3 $\pm$ 0.6	23.9 $\pm$ 0.6
	BRIDGE+SDCLR (ours)	37.1 $\pm$ 1.3	11.8 $\pm$ 0.9	1.3 $\pm$ 0.1	2.2 $\pm$ 0.2	6.8 $\pm$ 0.5	5.9 $\pm$ 0.7	12.5 $\pm$ 0.7	11.1 $\pm$ 0.6

F Full Results

G Algorithm

Table 14: Pretraining objective ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
SimCLR	71.8 ± 1.2	43.5 ± 0.5	9.0 ± 0.5	16.9 ± 0.3	27.2 ± 1.5	33.2 ± 3.3	41.7 ± 1.3	34.7 ± 1.2
MoCo	46.6 ± 3.1	18.4 ± 2.0	3.1 ± 0.5	4.3 ± 0.8	10.4 ± 3.2	15.0 ± 1.8	26.6 ± 1.9	17.8 ± 1.9
DINO	43.6 ± 0.7	17.8 ± 0.6	1.3 ± 0.5	1.6 ± 0.2	12.6 ± 4.4	8.7 ± 1.6	20.6 ± 1.1	15.2 ± 1.3

Table 15: Pretraining objective ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
SimCLR	66.3 ± 0.8	35.3 ± 0.4	5.1 ± 0.7	9.5 ± 0.1	24.8 ± 0.6	18.9 ± 1.0	36.7 ± 0.7	28.1 ± 0.6
MoCo	56.4 ± 1.6	25.9 ± 0.8	3.2 ± 0.4	5.2 ± 0.4	16.3 ± 1.1	12.2 ± 0.7	24.3 ± 0.7	20.5 ± 0.8
DINO	34.4 ± 0.2	12.4 ± 0.1	1.6 ± 0.1	2.4 ± 0.1	6.2 ± 0.8	5.4 ± 0.4	10.7 ± 0.8	10.4 ± 0.4

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Table 16: Augmentation mechanism ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Stable Diffusion 3	75.0 $\pm$ 0.2	48.4 $\pm$ 0.2	15.1 $\pm$ 1.2	20.6 $\pm$ 0.7	37.4 $\pm$ 2.0	41.2 $\pm$ 0.7	51.8 $\pm$ 1.1	41.4 $\pm$ 0.8
FLUX	70.7 $\pm$ 0.4	42.1 $\pm$ 1.4	10.3 $\pm$ 1.3	17.5 $\pm$ 0.9	29.4 $\pm$ 4.6	31.5 $\pm$ 2.9	43.7 $\pm$ 1.4	35.0 $\pm$ 1.9
Re-population	67.0 $\pm$ 2.6	39.4 $\pm$ 2.2	8.0 $\pm$ 0.6	15.4 $\pm$ 0.5	25.7 $\pm$ 1.4	31.7 $\pm$ 3.8	40.6 $\pm$ 0.5	32.6 $\pm$ 1.7

Table 17: Augmentation mechanism ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Stable Diffusion 3	68.3 $\pm$ 0.4	36.8 $\pm$ 0.2	6.5 $\pm$ 0.4	11.7 $\pm$ 0.6	35.1 $\pm$ 0.4	24.2 $\pm$ 0.3	45.9 $\pm$ 0.5	32.7 $\pm$ 0.4
FLUX	65.8 $\pm$ 1.0	35.4 $\pm$ 0.8	5.7 $\pm$ 0.7	9.4 $\pm$ 0.9	27.7 $\pm$ 1.0	18.8 $\pm$ 1.3	38.7 $\pm$ 1.2	28.8 $\pm$ 1.0
Re-population	64.5 $\pm$ 0.1	33.4 $\pm$ 0.6	4.2 $\pm$ 0.6	9.0 $\pm$ 0.4	23.0 $\pm$ 0.5	17.0 $\pm$ 0.4	34.2 $\pm$ 1.2	26.5 $\pm$ 0.5

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Table 18: Sample-selection strategy ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Mode window	71.9 ± 0.3	43.4 ± 0.7	9.0 ± 0.5	16.9 ± 0.2	27.9 ± 3.1	31.6 ± 3.3	41.7 ± 1.3	34.6 ± 1.3
Top tail	70.8 ± 0.5	43.1 ± 1.0	10.0 ± 1.7	15.9 ± 1.3	27.7 ± 1.0	30.5 ± 1.2	42.2 ± 1.4	34.3 ± 1.2
Uniform	71.5 ± 0.2	43.8 ± 0.7	9.3 ± 0.9	17.5 ± 1.2	32.9 ± 0.4	31.7 ± 3.9	42.0 ± 0.9	35.5 ± 1.2

Table 19: Sample-selection strategy ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Mode window	66.3 ± 0.8	35.3 ± 0.4	5.1 ± 0.7	9.5 ± 0.1	24.8 ± 0.6	18.9 ± 1.0	36.7 ± 0.7	28.1 ± 0.6
Top tail	65.7 ± 0.8	34.4 ± 0.2	4.1 ± 0.4	9.6 ± 0.2	23.9 ± 0.9	18.1 ± 0.8	35.4 ± 1.2	27.3 ± 0.6
Uniform	66.1 ± 1.0	35.7 ± 0.3	4.9 ± 0.1	10.2 ± 0.7	25.3 ± 0.4	19.0 ± 0.8	37.4 ± 0.6	28.4 ± 0.6

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Table 20: Cycle count ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
2 cycles	67.9 $\pm$ 0.6	40.7 $\pm$ 0.6	9.4 $\pm$ 1.8	16.0 $\pm$ 1.5	30.5 $\pm$ 3.2	27.4 $\pm$ 1.7	40.1 $\pm$ 0.4	33.1 $\pm$ 1.4
5 cycles	70.7 $\pm$ 1.7	43.3 $\pm$ 0.6	9.9 $\pm$ 1.5	16.4 $\pm$ 0.9	26.6 $\pm$ 2.7	31.2 $\pm$ 3.7	42.0 $\pm$ 1.2	34.3 $\pm$ 1.8
10 cycles	70.5 $\pm$ 1.2	43.8 $\pm$ 0.6	9.2 $\pm$ 0.9	17.2 $\pm$ 1.6	24.8 $\pm$ 5.2	31.3 $\pm$ 1.7	41.9 $\pm$ 0.6	34.1 $\pm$ 1.7
20 cycles	71.6 $\pm$ 1.1	44.6 $\pm$ 0.9	9.8 $\pm$ 0.3	17.3 $\pm$ 1.2	27.0 $\pm$ 2.1	33.3 $\pm$ 4.4	42.1 $\pm$ 1.5	35.1 $\pm$ 1.6

Table 21: Cycle count ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
2 cycles	65.1 $\pm$ 0.1	34.1 $\pm$ 0.5	4.6 $\pm$ 0.3	9.3 $\pm$ 0.3	23.8 $\pm$ 0.3	17.3 $\pm$ 0.7	34.5 $\pm$ 0.6	26.9 $\pm$ 0.4
5 cycles	66.0 $\pm$ 0.9	35.0 $\pm$ 0.9	5.0 $\pm$ 0.4	9.8 $\pm$ 0.2	24.8 $\pm$ 0.4	18.0 $\pm$ 0.5	35.6 $\pm$ 1.2	27.7 $\pm$ 0.6
10 cycles	66.4 $\pm$ 0.5	35.4 $\pm$ 0.3	4.8 $\pm$ 0.3	10.4 $\pm$ 0.6	25.2 $\pm$ 0.8	18.6 $\pm$ 0.9	36.0 $\pm$ 0.7	28.1 $\pm$ 0.6
20 cycles	66.5 $\pm$ 0.7	35.6 $\pm$ 0.8	4.8 $\pm$ 0.3	9.5 $\pm$ 0.4	24.9 $\pm$ 0.5	18.6 $\pm$ 1.2	36.3 $\pm$ 1.2	28.0 $\pm$ 0.7

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Table 22: Backbone architecture ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ResNet-18	67.7 $\pm$ 0.8	40.4 $\pm$ 0.5	10.6 $\pm$ 0.7	15.4 $\pm$ 0.3	26.1 $\pm$ 2.4	29.5 $\pm$ 2.7	41.0 $\pm$ 1.2	32.9 $\pm$ 1.2
ResNet-50	71.7 $\pm$ 1.0	44.0 $\pm$ 1.1	9.0 $\pm$ 0.5	17.4 $\pm$ 0.7	28.8 $\pm$ 1.7	32.0 $\pm$ 1.6	41.7 $\pm$ 1.3	34.9 $\pm$ 1.1
ViT-S	62.7 $\pm$ 0.4	37.6 $\pm$ 0.6	8.8 $\pm$ 0.5	12.2 $\pm$ 0.2	35.1 $\pm$ 4.2	30.2 $\pm$ 1.6	42.5 $\pm$ 1.2	32.7 $\pm$ 1.3
ViT-B	66.0 $\pm$ 2.3	40.6 $\pm$ 2.3	8.3 $\pm$ 0.5	14.3 $\pm$ 0.8	38.8 $\pm$ 2.6	31.3 $\pm$ 2.1	44.5 $\pm$ 1.7	34.8 $\pm$ 1.8

Table 23: Backbone architecture ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ResNet-18	65.0 $\pm$ 0.3	34.4 $\pm$ 0.3	4.7 $\pm$ 0.3	9.6 $\pm$ 0.4	23.5 $\pm$ 1.4	16.6 $\pm$ 0.3	36.3 $\pm$ 1.1	27.2 $\pm$ 0.6
ResNet-50	66.3 $\pm$ 0.8	35.3 $\pm$ 0.4	5.1 $\pm$ 0.7	9.5 $\pm$ 0.1	24.8 $\pm$ 0.6	18.9 $\pm$ 1.0	36.7 $\pm$ 0.7	28.1 $\pm$ 0.6
ViT-S	67.3 $\pm$ 0.4	37.5 $\pm$ 0.3	5.3 $\pm$ 0.2	7.6 $\pm$ 0.1	36.2 $\pm$ 0.7	20.1 $\pm$ 1.0	42.1 $\pm$ 1.0	30.9 $\pm$ 0.5
ViT-B	69.3 $\pm$ 0.8	40.1 $\pm$ 0.8	6.0 $\pm$ 0.2	8.3 $\pm$ 0.2	39.8 $\pm$ 0.8	21.9 $\pm$ 1.4	45.1 $\pm$ 0.7	32.9 $\pm$ 0.7

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Algorithm 1 BRIDGE training loop

```
1: Initialize unlabeled source set  $D^{(0)}$ , encoder  $f_\theta$ , frozen generator  $g_\phi$ 
2: for  $c = 0, \dots, C - 1$  do
3:   Pretrain  $f_\theta$  on  $D^{(c)}$  for  $E_c$  epochs
4:   Embed all images in  $D^{(c)}$  and compute mean  $k$ NN distances  $\{d_i^{(c)}\}$ 
5:   Clip the distance range, build a histogram, and find the modal bin center  $m^{(c)}$ 
6:   Select  $B$  images whose OOD distances are closest to  $m^{(c)}$ 
7:   Generate  $G$  synthetic images per selected image with frozen generator  $g_\phi$ 
8:   Form  $D^{(c+1)}$  by adding the generated images to  $D^{(c)}$ 
9: end for
```

557 Answer: [\[Yes\]](#)

558 Justification: The work uses existing public datasets and model releases under their pub-
559 lished terms, stays within academic research usage, and does not involve human subjects
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